Bartlett adjustment for random effects meta-analysis of event outcomes with small samples

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Abstract. Stijnen et al. (2010) describes four random effects models for meta-analyses of event outcomes. Inference for these models is based on an asymptotic chi-squared approximation. We describe a procedure for Bartlett correction of the confidence interval, improving the convergence rate from $1/n$ to $1/n^2$ under ideal conditions. We apply the proposed procedure to meta-analyses considered by Stijnen et al. (2010) on anti-infection devices, finding a differences in CI lengths up to 10%.

Keywords: Bartlett correction, generalized linear mixed models, meta-analysis

1 Introduction

There is currently underway a move to favor generalized linear mixed models in carrying out meta-analyses of event outcomes, particularly when sample sizes are small or events are rare. GLMMs avoid many of the dubious assumptions and devices, such as perfect knowledge of the study standard deviations and zero-cell corrections, required by the traditional alternative, inverse-variance weighting after a normalizing transformation. However, inferences under GLMMs are still based on an asymptotic chi-squared approximation. Therefore, as in the case of inverse-variance weighted estimation, the conclusions of smaller meta-analyses are less trustworthy, though smaller meta-analyses are where the use of GLMMs is often recommended. A classical remedy is Bartlett correction, which leverages modeling assumptions to improve the efficiency of the testing procedure. When the modeling assumptions are met, the improvement is typically a jump in the rate of asymptotic convergence from $1/n$ to $1/n^2$.

While Bartlett corrections have long been applied to generalized linear models (Cordeiro, 1983), there do not appear to be extensions to even the simple GLMMs encountered in meta-analysis. The reason for the reluctance appears to be computational difficulty (Hanada and Sugimoto, 2026). While computer algebra systems have frequently been applied to compute the third- and fourth-order derivative arrays needed for Bartlett correction of GLMs (Cordeiro, 1993; McCullagh, 1987; Andrews and Stafford, 2000), closed-form expressions for GLMM Bartlett corrections are unavailable for any of the commonly used models other than the normal-normal (Noma, 2011; Michael, 2026). A combination of automatic symbolic manipulation and numerical approximation appears necessary. We argue that with

proper design these difficulties can be overcome and consumer-grade hardware is sufficient for practical implementation.

In the next section we present four event data meta-analysis models, following Stijnen et al. (2010)'s exposition. We then summarize the method of Bartlett correction, and discuss some of the underlying assumptions. In Section 3 we offer a practical approach to estimation of the Bartlett correction applicable to the four models described in Section 2. In Section 4 we examine by simulation the effectiveness and robustness of the corrected test, finding that the proposed approximations are suitable for achieving the desired rate boost and identifying common model violations that may neutralize the advantage. In Section 5 we re-visit the data analyses of Stijnen et al. (2010). We find that, though the advantage of Bartlett correction may be negligible for meta-analyses of moderate or large size, it may be clinically relevant in smaller meta-analyses.

2 Background

2.1 Statistical models

We consider the random effects model for a meta-analysis consisting of n studies. In this model, a Gaussian random variable models the effects $\theta_1, \ldots, \theta_n$, of the individual studies in the meta-analysis,

$$\theta_i = \mu_0 + \tau_0 Z_i, \qquad Z_i \sim N(0,1)$$

The target of inference in the meta-analysis is μ_0, with τ_0^2 a nuisance parameter. The model for the conditional distribution of the data given the random effect depends on the type of analysis being conducted. Following Stijnen et al. (2010) we consider four types of analyses, determined by whether the studies included are single-arm or two-arm trials, and whether the estimand is based on proportions or incidence.

1. Single arm, proportions. Each study reports the number of events Y_i and total number of trials m_i. The statistical model is binomial

$$\begin{aligned} f_{m_i}(y_i \mid Z_i) &= \binom{m_i}{y_i} p^{y_i}(1-p)^{m_i - y_i} \\ p = p(\mu_0, \tau_0^2) &= \text{logit}^{-1}(\mu_0 + \tau_0 Z_i) \end{aligned} \tag{1}$$

   The target of inference μ_0 represents the log-odds of an event. The number of trials m_i is treated as a parameter of the conditional PMF f_{m_i}, which therefore typically varies across the studies in the meta-analysis.

2. Two arms, proportions. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the total number of trials $m_i^{(T)}$ and $m_i^{(C)}$. The model is conditional on the total number of events $S_i = Y_i^{(T)} + Y_i^{(C)}$, implying $Y_i^{(T)}$ follows

Fisher's non-central hypergeometric distribution

$$f_{m_i^{(T)},m_i^{(C)},S_i}(y_i^{(T)} \mid Z_i) = \frac{\binom{m_i^{(T)}}{y_i^{(T)}}\binom{m_i^{(C)}}{S_i - y_i^{(T)}}\omega^{y_i^{(T)}}}{\sum_{\max(0,S_i-m_i^{(C)})}^{\min(S_i,m_i^{(T)})}\binom{m_i^{(T)}}{y}\binom{m_i^{(C)}}{S_i-y}\omega^{y}} \qquad (2)$$
$$\omega = \omega(\mu_0, \tau_0^2) = \exp(\mu_0 + \tau_0 Z_i)$$

The target of inference μ_0 represents the ratio of the treatment to control log-odds. The parameters of the conditional PMF for study i are $m_i^{(T)}, m_i^{(C)}$, and S_i.

3. Single arm, incidence rate. Each study reports the number of events Y_i and the duration of observation T_i. The model is Poisson,

$$f_{T_i}(y_i \mid Z_i) = \exp(-\lambda)\frac{\lambda^{y_i}}{y_i!} \qquad (3)$$
$$\lambda = \lambda(\mu_0, \tau_0^2) = T_i \exp(\mu_0 + \tau_0 Z_i)$$

The target of inference μ_0 represents the log of the incidence rate. The follow-up time T_i is treated as a parameter of the conditional PMF.

4. Two arms, incidence rates. Each study reports, for treatment and control arm, the number of events $Y_i^{(T)}$ and $Y_i^{(C)}$ and the duration of observation $T_i^{(T)}$ and $T_i^{(C)}$. Conditionally on $S_i = Y_i^{(T)} + Y_i^{(C)}$, $Y_i^{(T)}$ is binomial,

$$f_{T_i^{(T)},T_i^{(C)},S_i}(y_i^{(T)} \mid Z_i) = \binom{S_i}{y_i^{(T)}} p^{y_i^{(T)}}(1-p)^{S_i - y_i^{(T)}} \qquad (4)$$
$$p = p(\mu_0, \tau_0^2) = \text{logit}^{-1}(\log(T_i^{(T)}/T_i^{(C)}) + \mu_0 + \tau_0 Z_i)$$

The target of inference is the logarithm of the ratio of the incidence rates between the two arms. The sum of events S_i and the periods $T_i^{(T)}$ and $T_i^{(C)}$ are teated as parameters of the conditional PMF.

In all cases the likelihood for study i has the form

$$L_i(\mu, \tau^2 | Y_i) = \int_{-\infty}^{\infty} f_i(Y_i|z)\phi(z)dz, \qquad i = 1, \ldots n \qquad (5)$$

where $f_i(y|z)$ is study i's PMF conditional on a Gaussian random effect. Furthermore, the meta-analysis parameters (μ, τ^2) only enter the likelihood (5) through a parameter of the conditional PMF, namely, the mean λ in the case of the Poisson, the success probability p in the cases of the binomial, or the odds ratio ω in the case of the non-central hypergeometric.

The likelihood ratio ("LR") test of the null $\mu = \mu_0$ is based on the test statistic

$$W = 2(l(\hat{\mu}, \hat{\tau}^2) - l(\mu_0, \hat{\tau}_0^2)) \qquad (6)$$

where

$$l(\mu, \tau^2) = \sum_{i=1}^{n} \log L_i(\mu, \tau^2 | Y_i) \qquad (7)$$

is the total log-likelihood, $(\hat{\mu}, \hat{\tau}^2)$ maximizes $(\mu, \tau^2) \mapsto l(\mu, \tau^2)$, and $\hat{\tau}_0^2 = \hat{\tau}_0^2(\mu_0)$ maximizes $\tau^2 \mapsto l(\mu_0, \tau^2)$. In routine applications W converges to a chi-squared distribution under the null hypothesis, by Wilks' Theorem. Relying on this asymptotic approximation a confidence interval for μ_0 can be formed by inverting the test statistic; see Figure 2 for an example.

2.2 Bartlett correction

Under a range of conditions the mean of a correctly specified LR statistic (6) admits an expansion in integral powers of $1/n$ (Barndorff-Nielsen and Cox, 1984)

$$\mathbb{E}(W) = 1 + b/n + O(1/n^2).$$

Therefore, the Bartlett-corrected statistic

$$W_B = \frac{W}{1 + b/n} \tag{8}$$

has a mean agreeing with that of a chi-squared random variable up to order $O(1/n^2)$. The results of Bartlett (1937); Lawley (1956) show further that, under appropriate conditions, all other moments, and so also the entire distribution, agree with a chi-squared up to order $O(1/n^2)$. This rate improvement holds even when MLEs are plugged in for any unknown parameters appearing in the bias term b of the expansion (Barndorff-Nielsen and Hall, 1988). As with the LR statistic, the adjusted statistic may be inverted to obtain CIs and p-values; see Figure 2 for an example.

Lawley (1956) provides a formula for the correction b appearing in (8) as sums and products of the components of the inverse Fisher information for (7) and certain tensors $k_r^{(s)}$. Being a small set of sums and products this formula does not introduce numerical or computational difficulties that aren't already present in the basic tensor components input into the formula, so we focus here on the tensor components. To this end, let D_r denote partial differentiation with respect to a sequence of the parameters μ and τ^2 given by r. For example, $D_{\mu,\tau^2} \sum_{i=1}^n \log L_i(\mu, \tau^2 \mid Y_i)$ denotes the off-diagonal term of the observed information matrix for (7). Define $k_r = k_r^{(0)} = \mathbb{E} D_r \sum_{i=1}^n \log L_i(\mu, \tau^2 \mid Y_i)$ and $k_r^{(s)} = D_s k_r$. In this notation, for example, the Fisher information matrix has components $k_{\mu,\mu}, k_{\mu,\tau^2}, k_{\tau^2,\mu}, k_{\tau^2,\tau^2}$. Lawley's formula for the Bartlett correction requires all terms k_r where r is of length 2, 3, or 4, and all terms $k_r^{(s)}$ where r is of length 3 and s of length 1 or r is of length 2 and s of length 1 or 2. Taking into account the commutativity of differentiation, there are 35 such terms in a model with one parameter of interest and one nuisance parameter.

By linearity of expectation and differentiation, it suffices to compute the terms $k_r^{(s)}$ for the unit log-likelihoods $\log L_i(\mu, \tau^2 \mid Y_i), i = 1, \ldots, n$. Let us consider a single study from the meta-analysis, dropping the subscript i from the conditional PMFs and likelihoods. Then the general tensor component needed for the computation of Bartlett's correction is

$$k_r^{(s)} = \sum_y D_s \left(\left(D_r \log \int f(y|z)\phi(z)dz \right) \int w_k f(y|z)\phi(z)dz \right) \tag{9}$$

The outer sum is taken over the support of Y.

2.3 Assumptions

Among the assumptions underlying the above development we focus on three prominent ones. The first two stem from assumptions of the basic LR tests described in Stijnen et al. (2010), while the third is specific to Bartlett correction.

1. The Bartlett correction requires correct specification of the likelihood. We refer to Stijnen et al. (2010) for other modeling assumptions implicit in models (1)–(4) and focus here on Gaussianity of the unobserved random effect. While the LR test itself depends on Gaussianity for valid inference, the Bartlett correction may be less robust to deviations.

2. Wilks' theorem, underlying the chi-squared approximation to the LR statistic, applies to points in the interior of the parameter space and thus excludes the fixed effects model $\tau_0^2 = 0$. It is possible the coverage rate of the LR test at a given sample size n appear unaffected by low heterogeneity, as has been suggested (Jackson et al., 2018; Bakbergenuly and Kulinskaya, 2018), while the more delicate rate analysis carried out here betrays an impact. After all, the LR test for τ^2 is known to be inconsistent at the boundary $\tau^2 = 0$ (Self and Liang, 1987) and the parameters (μ, τ^2) are not orthogonal in the models (1)–(4).

3. The arguments under which the Bartlett correction delivers a $1/n^2$ rate are usually justified by Cramér's condition for the underlying random variables, e.g., (Jensen, 1993). This condition excludes lattice variables, such as appear in models (1)–(4). Consequently, while Bartlett corrections have been derived for statistics based on lattice random variables since the 1980s (Cordeiro, 1983), their effectiveness does not rest on a firm theoretical foundation and in fact does not appear to be understood even in basic cases (Frydenberg and Jensen, 1989; Jensen, 1993; Götze and van Zwet, 2006).

We have no theoretical contributions to make here and will examine model robustness and the effect of discreteness by simulation.

3 Computing the tensor components $k_r^{(s)}$

3.1 Overview

To compute the tensor component (9), we approximate the integrals by a quadrature rule. Since all the likelihoods (1)–(4) involve integrating against a Gaussian, Gauss-Hermite ("GH") quadrature is a natural choice. This and similar methods (Liu and Pierce, 1994) provide weights $w_1, \ldots, w_{n_{GH}}$ and nodes $z_1, \ldots, z_{n_{GH}}$ in the domain of integration allowing the Gaussian integral of smooth integrands such as appear in (5) to be approximated by a weighted sum,

$$k_r^{(s)} \approx \sum_y D_s \left(\left(D_r \log \sum_j w_j f(y|z_j) \right) \sum_k w_k f(y|z_k) \right) \tag{10}$$

The weights and nodes of the quadrature may in general depend on the unknown parameters μ and τ^2. To simplify the above derivatives we assume the weights and nodes are constant with respect to the parameters, as discussed further in Section 3.2. The number of nodes n_{GH} controls the quality of the approximation and introduces a tuning parameter into the proposed method. As in (9), the outer sum is taken over the support of Y, which is a finite set in all the models considered here other than the Poisson-normal model (3). The Poisson model requires a truncation point for computation, which introduces a further tuning parameter.

Carrying out the differentiations in (10) yields rational expressions in derivatives of the weighted sums $\sum_j w_j f(y|z_j)$, say, $I_{kl}(y) = \sum_j w_j \frac{d^k}{d\mu^k} \frac{d^l}{d(\tau^2)^l} f(y|z_j)$. These 35 expressions for $k_r^{(s)}$ in terms of the I_{kl} are the same for any model based on a likelihood of the form (5) and may be pre-computed. For example, $k_{\mu,\mu} = \sum_y (I_{20}(y) - (I_{10}(y)^2/I_{00}(y)))$, and $k_{\mu,\tau^2}^{(\tau^2)} = \sum_y (I_{01}(y)^2 I_{10}(y)/I_{00}(y)^2 - (I_{02}(y)I_{10}(y) + I_{01}(y)I_{11}(y))/I_{00}(y) + I_{12}(y))$.

The derivatives $I_{jk}(y)$ of the model-specific conditional PMFs that appear in the 35 tensor components $k_r^{(s)}$ number 15: $I_{jk}(y)$ for $0 \le k + l \le 4$. The form of these derivatives may be derived independently of data but their specific values in general depend on parameters of the conditional PMF, e.g., the trial sizes m_i in the binomial-normal model (1), and must typically be computed for each study of a meta-analysis. However, since the conditional PMFs only depend on the model parameters (μ, τ^2) through the PMF's distributional parameters, the derivatives $I_{jk}(y), 0 \le k + l \le 4$, may be re-written as linear combinations of $I_{00}(y) = \sum_j w_j f(y|\theta_j)$ using standard recurrences (Johnson et al., 2005). The advantage of this form is that the terms $I_{00}(y)$ involve only the PMF itself, for which optimized and numerically stable routines are available, unlike the derivatives of the PMF. For example, two applications of the Poisson PMF recurrence $\frac{d}{d\lambda} f_\lambda(y \mid Z) = f_\lambda(y-1 \mid Z) - f_\lambda(y \mid Z), y \ge 1$, yield, for the Poisson-normal model (3),

$$\begin{aligned} I_{11}(y) &= \sum_j w_j \frac{d^2}{d\mu d\tau^2} f_{\lambda(\mu,\tau^2)}(y \mid z_j) \\ &= \frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2} I_{00}(y-2) + (\frac{d^2\lambda}{d\mu d\tau^2} - 2\frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2}) I_{00}(y-1) + \frac{d\lambda}{d\mu}\frac{d\lambda}{d\tau^2} I_{00}(y), \qquad y \ge 2 \end{aligned}$$

The approach in brief is then, for each of the n studies in the meta-analysis, 1. evaluate the 15 vectors $I_{jk}(y), y \in \text{supp}(Y)$, with $\text{supp}(Y)$ truncated in the case of Poisson Y, using only off-the-shelf routines for the conditional PMF (Poisson, binomial, or hypergeometric) and basic arithmetic, 2. compute the 35 $k_r^{(s)}$ terms using only the computed 15 $I_{jk}(y)$ vectors and basic arithmetic. The 15 $I_{jk}(y)$ terms can be computed in parallel, and once they are computed the 35 $k_r^{(s)}$ terms can be computed in parallel. As the studies are independent the entire computation of the study-level Bartlett correction can be parallelized across the n studies, as well.

3.2 Tuning parameters

We discuss the main choices made in applying the proposed method to a given dataset: the number of nodes to be used in the quadrature rule, whether the nodes should be chosen

using a fixed or adaptive rule, and, in the case of the Poisson-normal model, the truncation point for the support.

1. The main tuning parameter of the proposed method is n_{GH}, the number of nodes in the quadrature approximation. We propose selecting this parameter heuristically. In the proposed method there are 15 integrands to which quadrature is applied, i.e., the functions $z \mapsto \frac{d^k}{d\mu^k}\frac{d^l}{d(\tau^2)^l} f(y|z)\phi(z)$, $0 \leq k+l \leq 4$, which show up as the summands in $I_{kl}(y) = \sum_j w_j \frac{d^k}{d\mu^k}\frac{d^l}{d(\tau^2)^l} f(y|z_j)$. We therefore examine plots of the 15 integrands, weighted by the quadrature weights in place of the normal density, versus quadrature nodes. For a safe choice of n_{GH} the plotted values should be negligible outside the range of $z_1, \ldots, z_{n_{GH}}$. See Figure 3 for an example. Typical choices encountered in the datasets below, leading to apparent convergence up to .001 absolute error in the CI endpoints, are 100–200 nodes for fixed rule quadrature and 10–30 for adaptive rule quadrature.

2. The two main variants of GH quadrature are fixed and adaptive. Fixed GH uses the same n_{GH} nodes centered at 0 irrespective of the integrand, whereas adaptive GH transforms the nodes to take into account the mode and scale of the integrand. The drawback of adaptive quadrature is the cost of repeated numerical estimation of the mode and scale of the integrand of (5). For example, computing MLEs under models (1)–(4) by Newton-Raphson requires evaluating the integrals in (7) at a sequence of values (μ, τ^2), and under adaptive quadrature each such evaluation requires estimating the mode and scale of a new integrand, leading to nested optimizations. This cost is partially mitigated by the fewer nodes generally required by adaptive quadrature.

   A further difficulty with adaptive quadrature is conceptual. Since the adaptive nodes and weights depend on the parameters (μ, τ^2) through the mode and scale of the integrand, they are subject to the differentiation operators in (10), which is incompatible with the assumptions of the proposed method. Ignoring this dependence involves a tacit assumption that the mode and scale of the derivatives of the integrand are similar to the integrand's. However, this simplified approach leads to efficient implementation of adaptive GH. The only type of integrand for which a mode must then be computed is of the form (5). As all the conditional PMFs belong to the exponential family, these integrands are strictly log-concave and unimodal, and the mode and scale may be obtained quickly using Newton-Raphson.

   In most of the data analysis examples in Section 5, there is in fact little difference in performance between CIs returned by fixed and adaptive quadrature, the exception being the Poisson-normal analysis; see Figure 4.

3. Since the Poisson-normal model has infinite support, implementation of the proposed method requires a truncation point. We propose the use of visual heuristics to determine this tuning parameter, as well. The support of Y enters the calculations of the proposed method in the expression (10), where each of the 35 terms $k_r^{(s)}$ is approximated by a sum over y. These 35 summands give rise to 35 functions of y that may be plotted for a proposed truncation point. A satisfactory truncation point is a value beyond which the function values are negligible. See Figure 4. As the example there

shows, it may be difficult to know in general what is to be deemed negligible for a given data set, and moreover the truncation point often influences the choice of n_{GH}. For the Poisson-normal model we therefore suggest examining a matrix of CIs varying both parameters to identify the limiting CI. See Table 3 for an example.

4 Simulation

We examine by simulation the rate at which the LR and Bartlett-corrected LR statistics converge to the reference chi-squared distribution under the four models (1)–(4). The purpose is three-fold: 1. to investigate whether the quadrature approximations used by the proposed method are sufficient, 2. to investigate whether the Bartlett correction is effective as applied to this type of lattice data, as there are few theoretical results here, and 3. to assess the robustness of any improvement in the rate offered by Bartlett correction to deviations from the assumed non-degenerate Gaussian model for the random effect. Results are given in Table 1 and Figure 1.

4.1 Design

To make the simulation results relevant, the simulated meta-analyses are constructed to resemble the meta-analysis datasets of Niel-Weise et al. (2007) and Niel-Weise et al. (2008) analyzed by Stijnen et al. (2010) and re-analyzed in Section 5. The true effect μ_0 and heterogeneity τ_0^2 are taken to be the MLEs for the data obtained by Stijnen et al. (2010), and the distributional parameters are obtained by sampling with replacement from the datasets. For example, to simulate a meta-analysis of a single proportion of size n resembling the treatment arm data of Niel-Weise et al. (2007), we first sample with replacement n trial sizes $m_1, \ldots, m_n$, from the 18 treatment arm study sizes given by the corresponding column of Table 4; and second, sample n effects Y under the binomial likelihood (1) using the sampled trial sizes m_i and random success probabilities sampled using the parameters $(\mu_0, \tau_0^2) = (-3.50, .81)$ taken from the MLE calculations of Stijnen et al. (2010). Meta-analyses of sizes $10, \ldots, 20$, are sampled in this manner, reflecting the sizes $n = 9$ and $n = 18$ of Niel-Weise et al. (2007) and Niel-Weise et al. (2008). The rate of convergence of the empirical Kolmogorov-Smirnov distance to a chi-squared distribution is then estimated using n^4 Monte Carlo repetitions at each sample size n. See Michael (2026) for details on the rate estimation.

Robustness of the Gaussianity assumption is examined by perturbing a normal to a skewed distribution (Jones and Pewsey, 2009). The skew was chosen to test robustness since it drives the first-order error in CLT-type approximations. Robustness against non-degeneracy of the random effect, $\tau_0^2 > 0$, is studied in particular cases below.

In all simulations, we have used the true parameters (μ_0, τ_0^2), perturbed by a Gaussian with standard deviation $1/\sqrt{n}$, to calculate the Bartlett correction $b = b(\mu_0, \tau_0^2)$. In practice the MLE would be used. The practical reason is to avoid the prohibitive computational cost of re-calculating the Bartlett calculation for each of the n^4 Monte Carlo replicates. The justification is the result of Lawley (1956); Barndorff-Nielsen and Hall (1988) and others that the $O(1/n^2)$ Bartlett rate survives substitution of $\sqrt{n}-$consistent estimators into the Bartlett correction $b(\mu, \tau^2)$.

4.2 Results

Table 1 shows that when the model assumptions are met, the standard LR rate of $1/n$ and the Bartlett-corrected rate of $1/n^2$ are approximated by three of the four models (1)–(4). As skew increases, the rate deteriorates for both statistics. It appears also that skew affects the Bartlett-corrected statistic more severely than the standard statistic, with the corrected statistic in some instances showing a slower rate when skew is high. We therefore conclude that, at least in these three models and under correct specification of the random effect, the quadrature approximations may be sufficient.

The binomial-normal model for incidence rate ratios model (4) is the exception, showing a rate closer to $1/n^{3/2}$ than $1/n^2$ under Bartlett correction even in the idealized skew-free setting. This failure appears to be due to low heterogeneity. In addition to the MLE $\hat{\tau}^2 = 0.096$ of the original data set (Niel-Weise et al., 2008), we consider a smaller and larger choice of τ^2, maintaining other aspects of the simulation design. The results, presented in Fig. 1, show the rate deteriorating as τ^2 is decreased and improving to the target $1/n^2$ rate as τ^2 grows. The standard LR statistic does not appear much affected, consistent with previously reported simulations (Jackson et al., 2018; Bakbergenuly and Kulinskaya, 2018). We therefore conclude that the advantage of Bartlett correction can be impaired by low heterogeneity.

Table 1: Simulation results. The cells contain the observed medians and SDs of the log of the rate of convergence of the empirical Kolmogorov-Smirnov distance to a chi-squared distribution.

		Skew of random effect				
τ^2	**Statistic**	0	0.05	0.1	0.15	0.2
0.5	LR (original)	-0.78 (0.22)	-0.78 (0.15)	-0.67 (0.21)	-0.33 (0.08)	-0.11 (0.19)
	LR (Bartlett)	-1.27 (0.61)	-1.31 (0.37)	-1.13 (0.64)	-0.25 (0.32)	0.17 (0.92)
0.7	LR (original)	-1.01 (0.19)	-0.89 (0.17)	-0.66 (0.08)	-0.45 (0.11)	-0.14 (0.07)
	LR (Bartlett)	-1.72 (0.49)	-1.51 (0.33)	-0.51 (0.48)	-0.50 (0.45)	0.30 (0.09)
1.5	LR (original)	-1.36 (0.12)	-1.01 (0.21)	-0.87 (0.06)	-0.39 (0.13)	-0.08 (0.19)
	LR (Bartlett)	-1.92 (0.39)	-1.18 (0.24)	-0.81 (0.12)	-0.05 (0.20)	0.32 (0.19)

5 Data analysis

We revisit the example data analyses of Stijnen et al. (2010). Meta-analyses are carried out under the models (1)–(4) and 95% confidence intervals for the underlying effect μ_0 are calculated using the corrected and uncorrected profile LR. The results are presented in Table 2.

τ^2	**Statistic**	**Skew of random effect**				
		0	0.05	0.1	0.15	0.2
0.05	LR (original)	-1.04 (0.20)	-0.66 (0.34)	-1.31 (NA)	-0.85 (NA)	-0.63 (NA)
	LR (Bartlett)	-0.87 (0.24)	-0.84 (0.10)	-1.38 (NA)	-1.48 (NA)	-0.67 (NA)
0.12	LR (original)	-0.83 (0.21)	-0.73 (0.22)	-0.47 (0.23)	-0.14 (0.23)	0.07 (0.18)
	LR (Bartlett)	-1.34 (0.21)	-1.45 (0.34)	-1.78 (0.39)	-1.76 (0.62)	-0.01 (0.91)
0.30	LR (original)	-0.97 (0.13)	-0.79 (0.18)	-0.46 (0.06)	-0.14 (NA)	0.31 (NA)
	LR (Bartlett)	-2.44 (0.52)	-1.34 (0.44)	-0.30 (0.54)	0.55 (NA)	1.02 (NA)

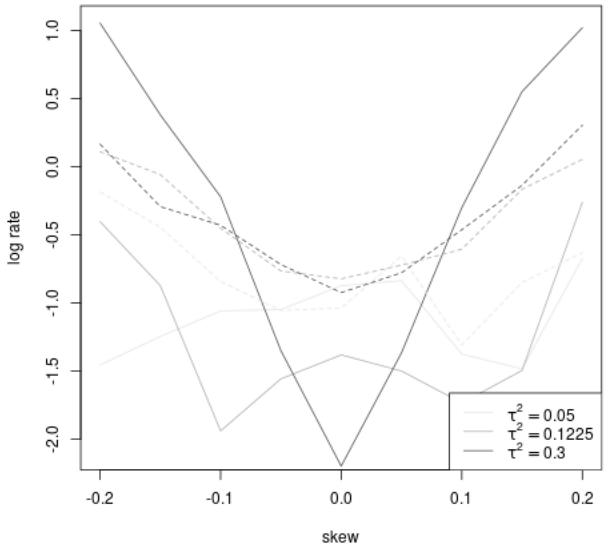


Figure 1: Further analysis of the simulation using the binomial model for IRR (4). The rate of the Bartlett-corrected LR statistic (solid line) drops from the target $1/n^2$ rate as the heterogeneity parameter τ^2 drop to 0 or as the skew of the random effect increases. The rate of the standard LR statistic (dashed lines) is also affected adversely by skew but does not appear to be affected by varying τ^2 in this range. The data is simulated based on Niel-Weise et al. (2008).

5.1 Proportions

Niel-Weise et al. (2007) carries out a meta-analysis on the efficacy of an anti-infection treatment for catheters. Each of the 18 included studies reports, for treatment and control arms, the total number of patients in the study and number of those who developed a catheter-related infection. See Table 4.

We apply the binomial-normal model for proportions (1) to obtain CIs for the log odds of infection in each arm separately and the hypergeometric-normal model (2) to obtain a CI for the log-odds ratio across the arms. In all three cases, the CI using the Bartlett-corrected profile likelihood is only slightly wider than the uncorrected, with the endpoints differing only in the hundredth's place or less. This agreement may reflect the moderate sample size of the the meta-analysis at 17–18 studies.

5.2 Incidence

Niel-Weise et al. (2008) presents a smaller meta-analysis on the efficacy of an anti-infection treatment for catheters. Each of the 9 included studies reports, for each arm, the number of catheter-related infections observed in a population as well as the total "catheter-days," the sum over the study population of the number of days each subject used a catheter.

We apply the Poisson-normal model (3) to obtain CIs for the log incidence rate in each arm, and the binomial-normal model for incidence rate ratios (4) to obtain a CI for the log-incidence-rate ratio. For this smaller meta-analysis the difference between the corrected and uncorrected CIs is more substantial, the corrected CIs being about 10% larger. These differences include a loss of significance at the nominal level $\alpha = .05$ for the log-IR ratio study. See Figure 2 for a plot of the profile LR used to obtain the CI for incidence under treatment.

The number of quadrature nodes used to approximate the Bartlett correction was selected by examining plots of the summands in $I_{kl}(y) = \sum_j w_j \frac{d^k}{d\mu^k} \frac{d^l}{d(\tau^2)^l} f(y|z_j)$, $0 \leq k + l \leq 4$, as discussed in Section 3.2. Since these integrands, as functions of z, appear to be supported within the range of $n_{GH} = 20$ quadrature nodes, this choice of the tuning parameter appears satisfactory. See Figure 3; such a matrix of plots is inspected for each of the studies in the dataset.

To obtain the truncation point for the support of the Poisson-normal model, the 35 summands in (10), corresponding to each of the 35 terms $k_r^{(s)}$, viewed as functions of y, were plotted against a proposed support, as discussed in 3.2. For the treatment group of Niel-Weise et al. (2008) with fixed quadrature Figure 4a shows that the summands are not convincingly contained in even a relatively large proposed support $0 \leq y \leq 3000$. On the other hand, when adaptive quadrature is used, 4b suggests the terms $k_r^{(s)}$ are negligible beyond the proposed truncation point of 350. This example shows that the two tuning parameters of the Poisson-normal model may need to be chosen in coordination. A matrix such as Table 3 may be of assistance. Taking $n_{GH} = 20$ adaptive nodes and $y_{max} = 1,500$ support points is sufficient for a CI with endpoints within 3/10000 of the converged value, where $n_{GH} = 100$ adaptive nodes and $y_{max} = 5000$ are assumed to produce the converged value.

Meta-analysis	Estimand	LR CI	Proposed CI	% Diff
N-W '07, control arm	log odds	[-4.07, -3.00]	[-4.09, -2.98]	+3.7%
N-W '07, treatment arm		[-5.70, -4.20]	[-5.73, -4.19]	+2.7%
N-W '07	log-odds ratio	[-2.21, -0.71]	[-2.25, -0.68]	+5.1%
N-W '08, control arm	log IR	[0.90, 1.90]	[0.85, 1.95]	+10.6%
N-W '08, treatment arm		[0.10, 1.55]	[0.02, 1.61]	+9.7%
N-W '08	log-IR ratio	[-1.05, -0.01]	[-1.12, 0.05]	+11.3%

Table 2: Standard and Bartlett-corrected confidence intervals for the example meta-analysis datasets considered by Stijnen et al. (2010). There appears to be little advantage in Bartlett correction for the meta-analyses based on Niel-Weise et al. (2007), which consists of 18 studies. There is a greater difference for the meta-analyses based on Niel-Weise et al. (2008), which has half as many studies.

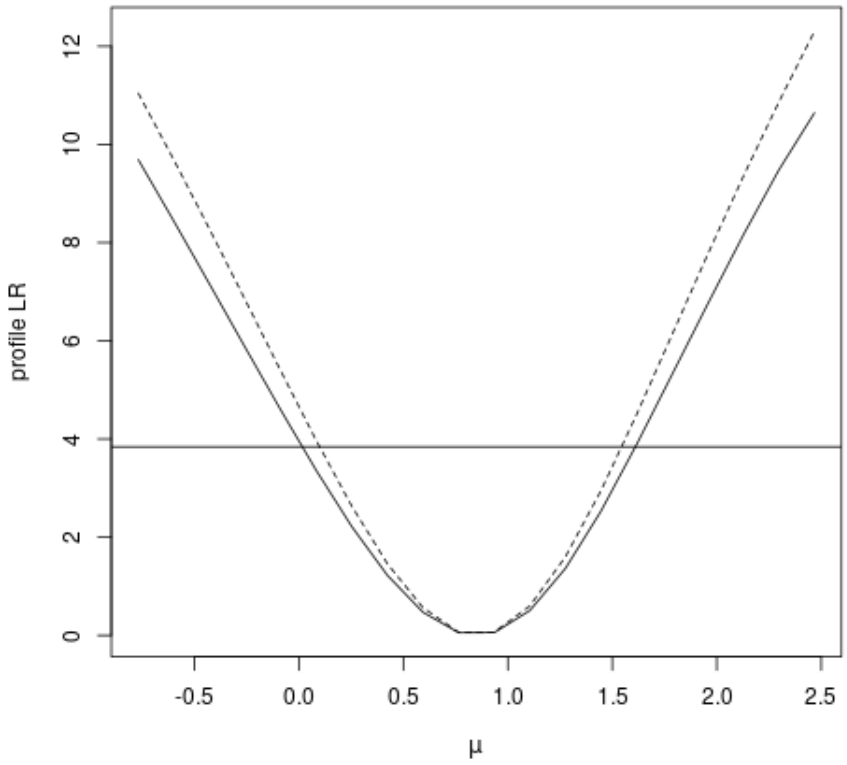


Figure 2: Profile curve for the standard (dashed line) and Bartlett-corrected (solid line) LR statistic. The data consists of the treatment arms of Niel-Weise et al. (2008).

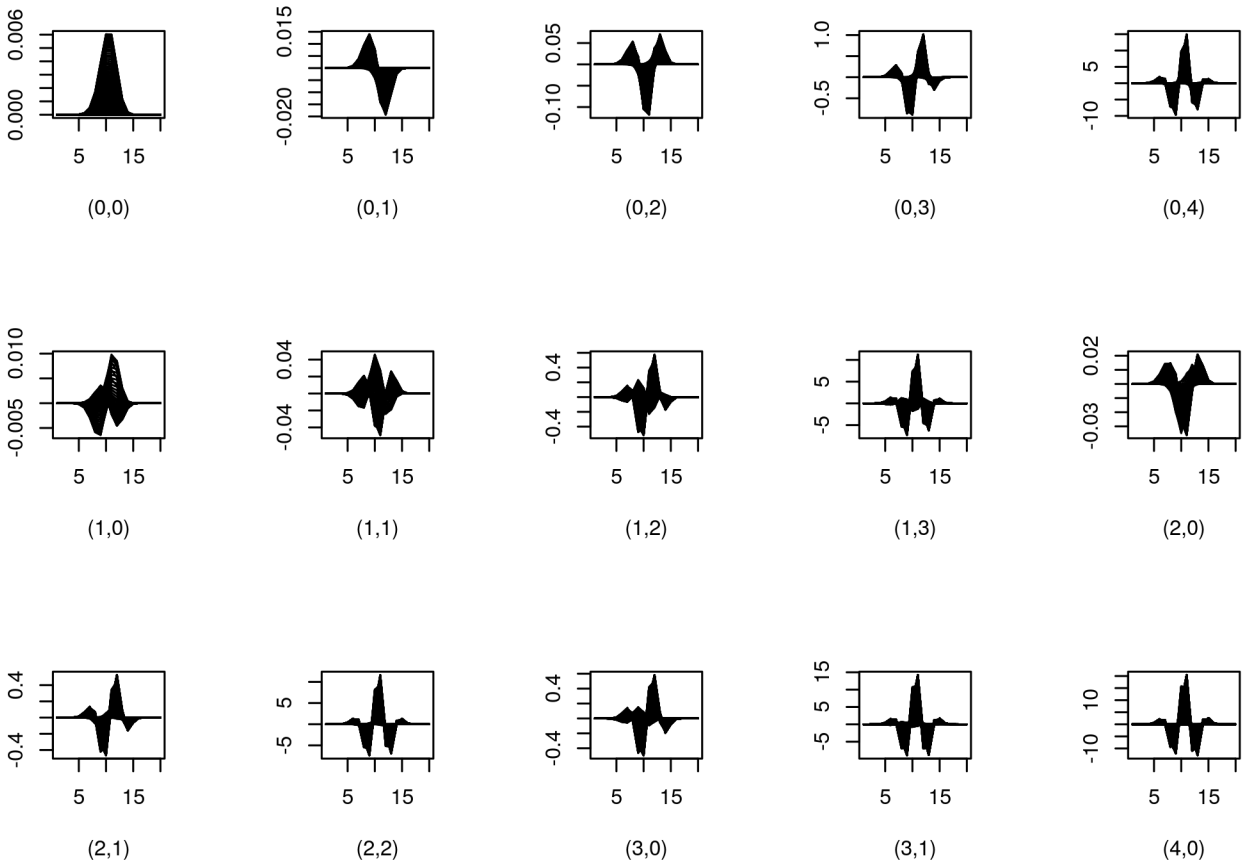
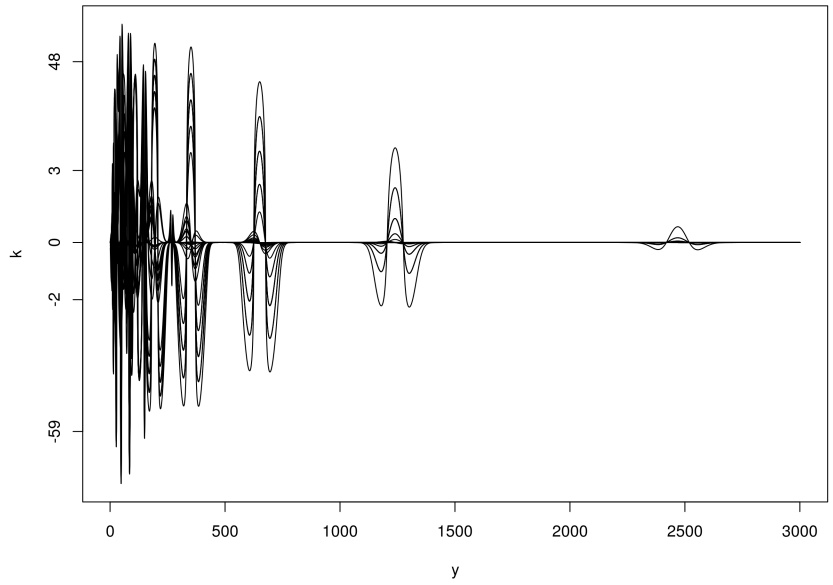


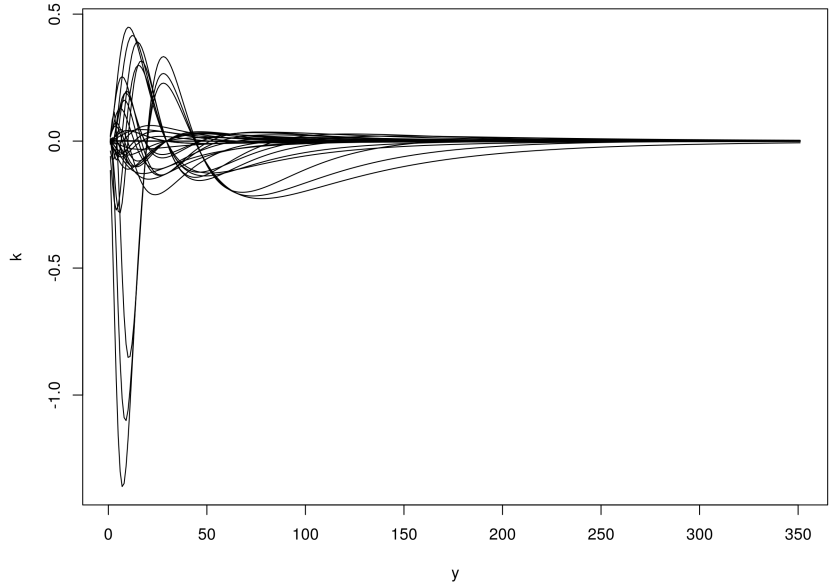
Figure 3: Diagnostic plot for determining the number of quadrature nodes n_{GH}. Corresponding to each of the 15 terms $I_{kl}(y), 0 \leq k + l \leq 4$, the integrand, viewed as a function of z, is plotted along proposed quadrature nodes, one path for each support point y. The depicted data is for study 3 of the treatment arm data of Niel-Weise et al. (2008).

	Poisson support cutoff $y_{\max}$		
$n_{\mathbf{GH}}$	**500**	**1000**	**1500**
10	[0.0219, 1.6102]	[0.0204, 1.6105]	[0.0202, 1.6113]
15	[0.0197, 1.6085]	[0.0181, 1.6091]	[0.0179, 1.6100]
20	[0.0200, 1.6083]	[0.0184, 1.6090]	[0.0182, 1.6098]

Table 3: Convergence of the Bartlett-corrected CI as the two tuning parameters of the Poisson-normal model are increased. The endpoints of the CI in the lower right cell, given by $n_{GH} = 20, y_{max} = 1.5 \times 10^3$, are each within 3×10^{-4} of the CI $(.0182745, 1.611421)$ obtained for $n_{GH} = 1 \times 10^2, y_{max} = 1 \times 10^5$.



(a) Non-adaptive quadrature (arcsinh-scaled).



(b) Adaptive quadrature.

Figure 4: Diagnostic plot for determining the truncation point for Y in the Poisson-normal model. The summands (10), viewed as functions of y, are plotted for a proposed support, one path for each of the 35 terms $k_r^{(s)}$. In the non-adaptive case 4a numerical artifacts appear up to support points as large as 3×10^3, and perhaps beyond, whereas for adaptive quadrature 4b the summands appear negligible beyond $y = 350$. A further concern in the case of non-adaptive quadrature is the magnitude of the terms. The depicted data is for study 3 of the treatment arm of the Niel-Weise et al. (2008).

6 Conclusion

We have presented a method to more accurately compute confidence intervals for meta-analyses of event outcomes. Discreteness of the data does not appear to be an obstacle to achieving fast rates. We have seen that the improvement offered by the proposed method deteriorates as the model assumption of a non-degenerate Gaussian random effect is progressively relaxed.

Ideally application of the Bartlett correction would be a "free lunch," offering an improved rate if the random effect is in fact Gaussian and otherwise being neutral, beyond perhaps the added computational cost. Since a Bartlett correction is only of order $O(1/n)$ whether misspecified or not, such a desire may not be unreasonable. We have seen this is not quite the case, however. When the skew of the random effect is sufficiently far from 0, the rate of the Bartlett-corrected statistic can be worse than the uncorrected statistic. Presumably a random effect sufficiently far from a Gaussian in other respects can also lead to this reversal.

Software implementing the proposed method and the routines used to carry out the simulations are available at the author's website.

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Study	$Y^{(C)}$	$n^{(C)}$	$Y^{(T)}$	$n^{(T)}$
1	3	117	0	116
2	3	35	1	44
3	9	195	2	208
4	7	136	0	130
5	6	157	5	151
6	4	139	1	98
7	3	177	1	174
8	2	39	1	74
9	19	103	1	97
10	2	122	1	113
11	7	64	0	66
12	1	58	0	70
13	5	175	3	188
14	11	180	6	187
15	0	105	0	118
16	1	262	0	252
17	3	362	1	345
18	1	69	4	64

Table 4: Data from Niel-Weise et al. (2007) analyzed in Section 5 under the binomial-normal model for proportions and the hypergeometric-normal model. Study 15 does not contribute to the hypergeometric-normal likelihood (2), reducing the sample size to 17 for that analysis in Section 5.

Study	$Y^{(C)}$	$T^{(C)}$	$Y^{(T)}$	$T^{(T)}$
1	11	1988	7	1344
2	8	1460	8	1599
3	14	10962	3	12012
4	10	1502	6	1536
5	1	482	1	370
6	8	913	1	729
7	15	6840	17	6760
8	7	1015	3	1107
9	3	440	2	320

Table 5: Data from Niel-Weise et al. (2008) analyzed in Section 5 under the binomial-normal model for incidence rate ratios and the Poisson-normal model.